

Generalized Board Games

From Tic-Tac-Toe to Chess as *xyz* Formal Systems

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Abstract

Generalized board games (GBGs) form an unusually concrete class of *xyz* formal systems. In the modern formulation used in *Depths of Induction and Formalized Self-Awareness, Version 48*, an *xyz* formal system $F = (x, y, z)$ has a nonempty alphabet x , a set y of well-formed finite strings, and a set z of finitary partial inference rules on y . A GBG specializes this framework by encoding rule-complete game states as well-formed formulas and legal moves as partial unary inference rules. A distinguished initial state acts as an axiom or initial hypothesis; finite legal reachability becomes theoremhood; a legal history becomes a sequential match-proof; and the match tree becomes the unfolding of the directed state graph.

Two examples show the same formalism at radically different scales. Tic-tac-toe is the small exact model: every legal move reduces the number of empty squares by one, every maximal play has at most nine plies, the full reachable state set and match tree can be enumerated exactly, and backward induction gives a draw from the empty board under perfect play. Chess is the large model: rule-complete states require richer metadata, shallow match-tree counts grow from 20 legal first moves to more than 6.9×10^{13} ten-ply histories, and Shannon's classical 10^{120} estimate appears naturally as a branching estimate for match-proofs. The paper also develops exact match recursions, state-versus-history projection, certified-state reasoning, alternating forced-win regions, and a forced-win horizon. Tic-tac-toe supplies transparency; chess supplies scale; both instantiate the same *xyz* proof-theoretic architecture.

1 Why Board Games Belong Inside the *xyz* Framework

The purpose of generalized board games is not to introduce a rival formalism beside *xyz* systems. The point is the opposite: board games provide a particularly visible family of *xyz* formal systems.

We use the modern definition from *Depths of Induction and Formalized Self-Awareness, Version 48* [2]. If y is a set, a k -ary rule on y is a partial function

$$r : y^k \rightharpoonup y,$$

where $k \geq 1$. A finitary rule is a k -ary rule for some positive finite k . An *xyz* formal system is an ordered triple

$$F = (x, y, z),$$

where x is a nonempty alphabet, y is a set of well-formed finite strings over x , and z is a set of finitary rules on y . The elements of z are the inference rules of the system.

This continues the earlier *xyz* framework of *Elements of Proof Theory*, where generalized board games already appeared as an example [1]. The modern partial-function formulation makes the intended role of legal moves especially clean.

The three coordinates may be remembered informally as

$$\boxed{x = \text{symbols}}, \quad \boxed{y = \text{well-formed formal objects}}, \quad \boxed{z = \text{licensed formal transitions}}.$$

A board game makes the third coordinate visible. A legal move is a formally licensed transition from one legal state to another. Thus, after a suitable finite encoding, a move can literally be an inference rule.

2 Rule-Complete States and Partial Move Rules

Let a game state contain whatever finite information is needed to determine all legal immediate continuations. For a board game one may begin with a location report

$$\ell : C \longrightarrow H \cup \{E\},$$

where C is the set of board locations, H is the set of piece or mark types, and E denotes emptiness. Additional rule-relevant information may be required.

Definition 2.1 (Rule-complete state). A state is *rule-complete* if it contains enough information to determine, from that state alone, whether every possible next move is legal, what successor state each legal move produces, and what rule-relevant terminal status results.

A rule-complete state may therefore have schematic form

$$s = (\ell, \tau, \eta),$$

where τ records the player to move and η contains any additional metadata. The tuple is understood through a fixed finite-string encoding in y .

Let $\mathcal{A}$ be a set of canonical concrete move labels. For each $\mu \in \mathcal{A}$, associate a partial unary rule

$$r_\mu : y \rightharpoonup y, \quad D_\mu = \text{dom}(r_\mu).$$

The condition $s \in D_\mu$ means exactly that move μ is legal at state s ; if so, $r_\mu(s)$ is the unique resulting state.

Definition 2.2 (GBG system). A *generalized board game* is an xyz formal system $F = (x, y, z)$ together with a distinguished initial state $s_0 \in y$, a set $\mathcal{A}$ of canonical move labels, and a bijection

$$\rho : \mathcal{A} \xrightarrow{\sim} z, \quad \rho(\mu) = r_\mu,$$

such that every element of y encodes a rule-complete game state and every r_μ is a partial unary rule describing exactly one legal move rule.

Thus

$$\boxed{\text{applicable inference rule} = \text{available legal move}}.$$

Partiality means only that a rule is not legal everywhere. It does not represent inaction. If a game permits passing, passing is represented by its own partial rule.

Definition 2.3 (Local finiteness). A GBG is *locally finite* if every state admits only finitely many legal moves:

$$|\{\mu \in \mathcal{A} : s \in D_\mu\}| < \infty \quad (s \in y).$$

Tic-tac-toe and ordinary chess are locally finite.

3 Theoremhood as Finite Reachability

Let the distinguished hypothesis set be

$$\Gamma = \{s_0\}.$$

Write

$$\text{Con}_F(\Gamma) = \{\varphi \in y : \Gamma \vdash_F \varphi\}$$

for the consequence set.

Theorem 3.1 (Reachability–Theorem Correspondence). *For every GBG,*

$$s \in \text{Con}_F(\{s_0\}) \iff s \text{ is reachable from } s_0 \text{ by finitely many, possibly zero, legal moves.}$$

Proof. A finite legal history

$$s_0 \rightarrow s_1 \rightarrow \cdots \rightarrow s_n = s$$

is a finite chain of applications of unary rules from z , hence a derivation of s from s_0 . Conversely, every inference rule of the GBG is a legal-move rule, so the dependency chain in a derivation from the single hypothesis s_0 is a finite legal state-transition path beginning at s_0 . $\square$

Hence, relative to the distinguished initial state,

$$\boxed{\text{reachable state} = \text{theorem}}.$$

4 Match-Proofs, Match Trees, and State Graphs

A match carries stricter temporal structure than an arbitrary proof: each next state must follow from the immediately preceding state.

Definition 4.1 (Match-proof). An n -ply match-proof beginning at s_0 consists of states

$$(s_0, s_1, \dots, s_n)$$

and move labels

$$(\mu_0, \dots, \mu_{n-1})$$

such that

$$s_{i+1} = r_{\mu_i}(s_i) \quad (0 \leq i < n).$$

Theorem 4.2 (Match-Proof Correspondence). *Legal move-labeled matches beginning at s_0 are in one-to-one correspondence with move-labeled match-proofs beginning at s_0 .*

Proof. A legal match records its states and move labels and therefore determines a match-proof. Conversely, the defining equations of a match-proof state that every consecutive transition is legal, so they recover a legal match. The two constructions are inverse. $\square$

Definition 4.3 (Match tree). The *match tree* $\mathcal{T}(s_0)$ is the rooted tree whose depth- n nodes are the n -ply move-labeled match-proofs beginning at s_0 .

The move-labeled transition graph has vertex set y and edge (s, μ, t) whenever $t = r_\mu(s)$. The associated state graph forgets the move label. Its reachable subgraph from s_0 has vertex set exactly $\text{Con}_F(\{s_0\})$.

Let $R_n(s_0)$ be the set of states reachable in exactly n plies, and let $M_n(s_0)$ be the number of move-labeled match-proofs of exactly n plies.

Proposition 4.4 (History-to-State Projection). *The endpoint map from the depth- n level of $\mathcal{T}(s_0)$ onto $R_n(s_0)$ is surjective. Hence, whenever $M_n(s_0)$ is finite,*

$$\boxed{|R_n(s_0)| \leq M_n(s_0).}$$

Different histories may therefore prove the same theorem state. In board-game language this is transposition. Equality of states means equality of *rule-complete* states, not merely equality of visible board diagrams.

5 Certified States

Suppose a state s is supplied together with a certificate that

$$s \in \text{Con}_F(\{s_0\}).$$

The certificate establishes that the state arose legally. Rule-completeness then allows future analysis to begin directly at s .

Proposition 5.1 (Certified-State Principle). *If two legal match-proofs terminate at the same rule-complete state s , then they have exactly the same legal immediate continuations and the same successor state for each applicable move label.*

Proof. Applicability and successor values are functions of the rule-complete state s itself. They do not depend on which proof of s was supplied. $\square$

6 Local Branching and Exact Match Counting

Define

$$\begin{aligned} \text{Succ}(s) &= \{r_\mu(s) : s \in D_\mu\}, \\ b(s) &= |\text{Succ}(s)|, \quad b_{\text{move}}(s) = |\{\mu : s \in D_\mu\}|. \end{aligned}$$

Then

$$b(s) \leq b_{\text{move}}(s),$$

with equality whenever different legal moves always produce different successor states.

Let $\mathcal{M}_n(s)$ be the set of move-labeled legal matches of exactly n plies beginning at s , and put

$$M_n(s) = |\mathcal{M}_n(s)|, \quad M_0(s) = 1.$$

Theorem 6.1 (Exact Match Recursion). *For every $n \geq 0$,*

$$\boxed{M_{n+1}(s) = \sum_{\mu: s \in D_\mu} M_n(r_\mu(s)).}$$

The sum is a cardinal sum in general and an ordinary finite sum for locally finite games.

Proof. Partition the $(n+1)$ -ply histories according to their unique first move μ . After that move, the remaining history is an n -ply history beginning at $r_\mu(s)$. $\square$

Corollary 6.2 (Uniform Branching Bound). *If every state reachable before depth n has at most B legal moves, then*

$$\boxed{M_n(s_0) \leq B^n.}$$

Corollary 6.3 (Layerwise Branching Bound). *If every state reachable after exactly j plies has at most B_j legal moves for $0 \leq j < n$, then*

$$M_n(s_0) \leq \prod_{j=0}^{n-1} B_j.$$

7 Forced Winning Regions and a Settlement Horizon

Let a two-player alternating GBG have players P and Q . Partition its nonterminal states into P -to-move and Q -to-move states. Let W_P be the terminal states in which P wins. A maximal play is either an infinite legal history or a finite history ending in a terminal state.

Define

$$A_0 = W_P.$$

Inductively let

$$A_{n+1} = A_n \cup E_n \cup U_n,$$

where E_n contains the nonterminal P -to-move states s satisfying

$$\exists t \in \text{Succ}(s) \quad t \in A_n,$$

and U_n contains the nonterminal Q -to-move states s satisfying

$$\text{Succ}(s) \neq \emptyset \quad \text{and} \quad \forall t \in \text{Succ}(s), t \in A_n.$$

Set

$$W_P^* = \bigcup_{n=0}^{\infty} A_n.$$

Theorem 7.1 (Finite Forced-Win Characterization). *In a locally finite two-player alternating GBG,*

$$s \in W_P^*$$

if and only if P has a strategy from s under which every maximal compatible play reaches W_P after finitely many plies.

Proof. If $s \in A_n$ for least n , the defining existential/universal clauses let P force the stage index to decrease until $A_0 = W_P$ is reached. Conversely, suppose P has a strategy under which every maximal compatible play reaches W_P in finite time. The strategy tree is finitely branching. If it had unbounded finite depth, Koenig's lemma would produce an infinite compatible branch, contradicting the hypothesis. Hence the strategy tree has finite height, and backward induction places its root in some A_N . $\square$

For $s \in W_P^*$ define the *forced-win horizon*

$$h_P(s) = \min\{n : s \in A_n\}.$$

Then $h_P(s) = 0$ on W_P , while at nonterminal forced-win states

$$h_P(s) = 1 + \min_{t \in \text{Succ}(s) \cap W_P^*} h_P(t)$$

when it is P 's turn, and

$$h_P(s) = 1 + \max_{t \in \text{Succ}(s)} h_P(t)$$

when it is Q 's turn.

This horizon differs from ordinary shortest-proof length: a shortest proof chooses one successful derivation, whereas a forced-win horizon must survive the opponent's choices.

8 Tic-Tac-Toe: The Small Canonical GBG

Tic-tac-toe is the most transparent nontrivial GBG example. It has enough structure to distinguish reachability from strategy, history trees from state graphs, and theoremhood from forced winning, while remaining exactly enumerable.

8.1 Encoding

Let the board locations be

$$C = \{1, 2, \dots, 9\},$$

and let a board report be

$$\ell : C \rightarrow \{X, O, E\}.$$

A rule-complete state records the board, whose turn it is when the game is ongoing, and whether the state is ongoing, an X win, an O win, or a draw. On reachable states the turn is already determined by the numbers of X and O marks, but recording it explicitly is harmless and keeps the presentation uniform.

A particularly economical canonical move set is

$$\mathcal{A}_{\text{TTT}} = \{1, 2, \dots, 9\},$$

where move label i means “place the current player's mark in square i .” The partial rule

$$r_i : y \rightarrow y$$

is defined exactly when square i is empty and the current state is nonterminal. Its output fills square i with the current player's mark, changes the player to move when appropriate, and records a terminal result if the new board ends the game.

Thus tic-tac-toe is an xyz formal system with nine canonical move rules.

8.2 A strict descent function

Let

$$e(s) = \text{the number of empty squares in } s.$$

Proposition 8.1 (Tic-Tac-Toe Finiteness). *For every legal tic-tac-toe move $s \rightarrow t$,*

$$\boxed{e(t) = e(s) - 1.}$$

Consequently every maximal tic-tac-toe match is finite, has at most nine plies, and the reachable state graph is acyclic.

Proof. Every legal move fills exactly one previously empty square and never empties a square. Since $e(s_0) = 9$ and $e(s) \geq 0$, no legal history can have more than nine moves. A directed cycle would force e to decrease and return to its previous value, which is impossible. $\square$

This argument is stronger than local finiteness: it gives a global finite horizon for the entire game.

8.3 Exact history and theorem-state counts

Because the game is so small, the exact match recursion can be evaluated over the entire reachable graph. The following counts use ordinary 3×3 tic-tac-toe, stop play immediately when a win occurs, distinguish board orientations, and do not quotient rotations or reflections.

Depth n	Match-proofs $M_n(s_0)$	Distinct reachable states $ R_n(s_0) $
0	1	1
1	9	9
2	72	72
3	504	252
4	3,024	756
5	15,120	1,260
6	54,720	1,520
7	148,176	1,140
8	200,448	390
9	127,872	78

The state counts sum to

$$\boxed{5,478}$$

reachable theorem states. The strict inequality

$$|R_n(s_0)| < M_n(s_0)$$

already appears at depth 3: different move orders can produce the same rule-complete state. Tic-tac-toe therefore exhibits transposition long before chess-scale complexity appears.

Since no win is possible before ply 5, the falling-factorial count remains exact through depth 5:

$$M_n(s_0) = \frac{9!}{(9-n)!} \quad (0 \leq n \leq 5).$$

Beginning at depth 6, wins occurring on ply 5 prune continuations, so the falling-factorial count becomes only an upper bound.

The complete maximal-history count is

$$\boxed{255,168}.$$

More precisely:

Terminal ply	X wins	O wins	Draws
5	1,440	0	0
6	0	5,328	0
7	47,952	0	0
8	0	72,576	0
9	81,792	0	46,080

These numbers are exact evaluations of the match recursion with terminal branches assigned no successors.

8.4 Reachable winning theorems are not forced wins

There are many reachable X -winning theorem states and many reachable O -winning theorem states. That alone does not tell us the value of the initial state.

For the finite game define a minimax value

$$v(s) \in \{-1, 0, 1\}$$

by assigning 1 to terminal X wins, -1 to terminal O wins, and 0 to terminal draws. For nonterminal states define

$$v(s) = \max_{t \in \text{Succ}(s)} v(t)$$

when X is to move, and

$$v(s) = \min_{t \in \text{Succ}(s)} v(t)$$

when O is to move.

The descent function $e(s)$ makes this recursion well-founded.

Proposition 8.2 (Perfect-Play Value of Tic-Tac-Toe). *For the empty tic-tac-toe board s_0 ,*

$$\boxed{v(s_0) = 0.}$$

Hence perfect play yields a draw: neither player has a forced win from the initial state.

Proof. Evaluate the displayed recurrence backward from terminal states in increasing order of the number of empty squares. This is a finite exact backward induction over the 5,478 reachable states. Equivalently, one may enumerate the reachable state graph, assign the terminal values 1, 0, and -1 , and propagate maxima at X -to-move states and minima at O -to-move states until the empty board is reached. This exhaustive procedure is deterministic and directly reproducible from the stated rules; it yields $v(s_0) = 0$. $\square$

Thus tic-tac-toe gives a particularly clean instance of

$$\boxed{\text{reachable winning theorem} \neq \text{forced win.}}$$

The game contains proofs of X -winning states, but X cannot force every opponent response to remain on such a proof path.

9 Chess: The Large-Scale Canonical GBG

Chess exhibits the same formal architecture but at a dramatically larger scale. A rule-complete chess state contains the board position and side to move together with rule-relevant metadata such as castling rights, en-passant status, move counters, and whatever repetition information the chosen rules convention requires.

Canonical concrete move labels may include, for example,

$$e2 - e4, \quad Ng1 - f3, \quad O - O,$$

with distinct labels for different promotion choices. Each label indexes a partial unary rule on the set of rule-complete chess states.

Hence, exactly as in tic-tac-toe,

legal chess move = application of a partial inference rule.

9.1 Finiteness under standard FIDE board rules

For the strategic interpretation of standard FIDE chess, the rule-complete state carries the information needed to implement the official draw rules. A draw may be claimed after 50 moves by each player without a pawn move or capture, while 75 such moves automatically draw the game; fivefold repetition is also automatic [6]. A claimable draw may be represented as a legal game-ending action, while an automatic draw is represented in the terminal status of the resulting state.

Proposition 9.1 (Finiteness of Standard FIDE Chess Matches). *In a GBG encoding of standard FIDE chess in which the official automatic draw provisions are represented, every maximal board-rule chess match is finite.*

Proof. Only finitely many pawn moves can occur: each pawn advances in one direction and eventually either disappears or promotes. Only finitely many captures can occur because the game starts with finitely many pieces and promotion replaces a pawn rather than creating an additional independent piece. Hence every play has a last pawn move or capture. If the game has not already ended, Article 9.6.2 of the FIDE Laws then forces a draw after 75 moves by each player without a pawn move or capture, with checkmate on the last move taking precedence [6]. Thus no infinite maximal play is possible. $\square$

Resignation and clock forfeiture can be added as extra terminal actions in a tournament model, but they are not needed for the finiteness argument above.

9.2 Exact shallow match-tree counts

In computer chess, *perft* counts legal move-generator paths from a position to a fixed depth. Under the standard perft convention of the cited enumeration, these are GBG move-labeled histories. From the standard initial position [4]:

Depth n (plies)	Exact $M_n(s_0)$
0	1
1	20
2	400
3	8,902
4	197,281
5	4,865,609
6	119,060,324
7	3,195,901,860
8	84,998,978,956
9	2,439,530,234,167
10	69,352,859,712,417

The first theorem layer contains only 20 one-ply match-proofs. Ten plies later the match tree already contains more than 6.9×10^{13} legal histories at that depth.

These are histories, not necessarily distinct theorem states. The general projection theorem still gives

$$|R_n(s_0)| \leq M_n(s_0).$$

9.3 Layerwise branching

OEIS A278830 records the following maximal legal-move counts attainable at the first ten chess plies [5]:

$$20, 20, 31, 32, 46, 48, 52, 55, 61, 63.$$

Thus

$$\begin{aligned} M_{10}(s_0) &\leq 20 \cdot 20 \cdot 31 \cdot 32 \cdot 46 \cdot 48 \cdot 52 \cdot 55 \cdot 61 \cdot 63 \\ &= 9,629,575,667,712,000. \end{aligned}$$

The exact value is

$$69,352,859,712,417.$$

The gap measures the slack introduced when a maximum available somewhere on a layer is used as though it were attained on every history.

9.4 Shannon scale

Shannon’s 1950 analysis used roughly 30 legal moves per position and roughly 40 moves per game, giving about 10^3 possibilities per White–Black move pair and hence

$$(10^3)^{40} = 10^{120}$$

possible variations at the resulting scale [3]. In GBG language this is immediately recognizable as an approximate branching calculation for match-proofs.

Using 30 branches for 80 plies gives

$$30^{80} \approx 1.48 \times 10^{118},$$

while Shannon’s convenient approximation $30^2 \approx 10^3$ produces 10^{120} . Neither figure is an exact game-tree count; both are scale estimates.

Shannon also discussed a rough static arrangement count on the order of

$$\frac{64!}{32!(8!)^2(2!)^6} \approx 10^{43}.$$

The contrast highlights a basic GBG distinction:

match-tree nodes count histories, reachable state-graph vertices count theorem states.

Many histories may merge into the same rule-complete state.

10 Tic-Tac-Toe and Chess Side by Side

The two examples occupy opposite ends of the same framework.

Feature	Tic-tac-toe	Chess
State data	Board, turn, terminal status	Board, turn, castling, en-passant, counters, repetition data as required
Canonical move rules	Nine square-placement rules	Concrete legal chess move rules
Local finiteness	Immediate	Immediate
Global play finiteness	At most 9 plies	Finite under standard FIDE automatic-draw rules
Reachable states	Exactly 5,478	Vast; no comparable small enumeration used here
Complete histories	Exactly 255,168	Enormous; Shannon-scale estimates illustrate growth
Perfect-play initial value	Draw	Not used as a theorem here
Pedagogical role	Transparency and exact enumeration	Scale and realistic complexity

The point is not that the two games have similar complexity. They plainly do not. The point is that their formal anatomy is the same:

$$\boxed{\text{state} \in y}, \quad \boxed{\text{move} \in z}, \quad \boxed{\text{reachable state} = \text{theorem}}.$$

Tic-tac-toe shows the architecture with essentially no combinatorial fog. Chess shows that the same architecture remains meaningful when the match tree becomes enormous.

11 Relation to General Proof Counting

The proof-counting development in *Depths of Induction and Formalized Self-Awareness, Version 48* defines $P_F(\Gamma, \varphi, n)$ as the number of proofs of a specified theorem φ having exactly n proof lines. If the maximum rule arity is a and

$$\kappa_k = |z_k(F)|,$$

then the general upper bound

$$P_F(\Gamma, \varphi, n) \leq \prod_{j=1}^n \left(|\Gamma| + \sum_{k=1}^a \kappa_k (j-1)^k \right)$$

is available [2].

GBG match-proofs are more temporally constrained. The premise of the next unary rule is fixed to be the immediately preceding state, so match counting is directed-path counting rather than arbitrary reuse of earlier proof lines. This explains why the exact GBG recursion is much sharper for game histories.

For a specified endpoint φ , define

$$M_n(s_0, \varphi) = |\{\pi \in \mathcal{M}_n(s_0) : \text{end}(\pi) = \varphi\}|.$$

Then, when the level is finite,

$$M_n(s_0) = \sum_{\varphi \in R_n(s_0)} M_n(s_0, \varphi).$$

This is the natural GBG analogue of endpoint-sensitive proof counting.

12 What Is General and What Is Game-Specific

The following statements arise from the GBG formalism itself:

$\text{Con}_F(\{s_0\})$ = the set of states finitely reachable from s_0 ;

$$M_{n+1}(s) = \sum_{\mu: s \in D_\mu} M_n(r_\mu(s));$$

$$|R_n(s_0)| \leq M_n(s_0);$$

the uniform and layerwise branching bounds; the match-tree/state-graph distinction; the certified-state principle; and, under local finiteness, the alternating characterization of finite forced wins.

By contrast, the nine-ply cap, the 5,478 reachable tic-tac-toe states, the 255,168 complete tic-tac-toe histories, the perfect-play draw value, chess perft values, FIDE-specific termination rules, and Shannon’s numerical estimates belong to the particular games and rule conventions being modeled.

This separation is important. A general formalism should expose where game-specific parameters enter without pretending that the same parameters hold for every game.

13 Conclusion

Generalized board games are concrete *xyz* formal systems. Their basic dictionary is

rule-complete game state = well-formed formal object,

legal move = partial unary inference rule,

finite reachability = theoremhood,

legal history = match-proof,

match tree = tree of sequential match-proofs,

and

strategy = adversarially controlled reachability.

Tic-tac-toe belongs first because it makes every part of this dictionary visible. Its strict descent function proves global finiteness, its state graph and match tree can be counted exactly, and its perfect-play draw cleanly separates reachable winning theorems from forced winning. Chess belongs next because it shows that the same formal language survives an enormous increase in combinatorial scale. The progression is therefore deliberate:

tic-tac-toe: transparency $\longrightarrow$ chess: scale.

The underlying lesson is simple. A formal system specifies what may legally follow from what. In proof theory that licensed step is called an inference. In a board game it is called a move. GBGs study those two descriptions as instances of the same mathematical structure.

References

- [1] Brian Tenneson, *Elements of Proof Theory*, Version 2.36, January 30, 2023.
- [2] Brian Tenneson, *Depths of Induction and Formalized Self-Awareness*, Version 48, 2026. Modern definition of finitary partial rules and *xyz* formal systems. Available at https://btenneson.github.io/pub/cs.L0_Logic_in_Computer_Science/depths_of_induction_v48/.
- [3] Claude E. Shannon, “Programming a Computer for Playing Chess,” *Philosophical Magazine*, Series 7, Vol. 41, No. 314, pp. 256–275, March 1950.
- [4] Chess Programming Wiki, “Initial Position Summary,” exact perft move-path enumerations for the standard initial chess position, accessed September 21, 2026, https://www.chessprogramming.org/Initial_Position_Summary.
- [5] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A278830, “Maximal number of possible moves at the n -th ply of a chess game,” accessed September 21, 2026, <https://oeis.org/A278830>.
- [6] FIDE, *FIDE Laws of Chess taking effect from 1 January 2023*, Articles 9.2, 9.3, and 9.6, FIDE Handbook, accessed September 21, 2026, <https://handbook.fide.com/chapter/e012023>.